# Lab Assignment-2

## Wireshark Filters

### Applying Filters

### TCP Filter

tcp

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 13 | 10.392455227 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 94 | 37660 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=233609373 TSecr=0 WS=512 |
| 14 | 10.392645999 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 94 | 37664 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=233609373 TSecr=0 WS=512 |
| 15 | 10.402946384 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 94 | 443 → 37660 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM TSval=3691152451 TSecr=233609373 WS=512 |
| 16 | 10.402946945 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 94 | 443 → 37664 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM TSval=2766903359 TSecr=233609373 WS=512 |
| 17 | 10.402991428 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 86 | 37660 → 443 [ACK] Seq=1 Ack=1 Win=65024 Len=0 TSval=233609384 TSecr=3691152451 |
| 18 | 10.403011000 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 86 | 37664 → 443 [ACK] Seq=1 Ack=1 Win=65024 Len=0 TSval=233609384 TSecr=2766903359 |
| 19 | 10.403174931 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 308 | Client Hello (SNI=firefox.settings.services.mozilla.com) |
| 20 | 10.403378805 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 308 | Client Hello (SNI=firefox.settings.services.mozilla.com) |
| 21 | 10.412344483 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 86 | 443 → 37664 [ACK] Seq=1 Ack=223 Win=218624 Len=0 TSval=2766903369 TSecr=233609384 |
| 22 | 10.417170689 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TLSv1.2 | 290 | [TCP Previous segment not captured] , Ignored Unknown Record |
| 23 | 10.417172973 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 86 | 443 → 37660 [ACK] Seq=1 Ack=223 Win=218624 Len=0 TSval=3691152462 TSecr=233609384 |
| 24 | 10.417173023 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TLSv1.2 | 2942 | Server Hello, Certificate |
| 25 | 10.417173073 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TLSv1.2 | 290 | Server Key Exchange, Server Hello Done |
| 26 | 10.417336811 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 98 | [TCP Dup ACK 18#1] 37664 → 443 [ACK] Seq=223 Ack=1 Win=65024 Len=0 TSval=233609398 TSecr=2766903369 SLE=2857 SRE=3061 |
| 27 | 10.417856270 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 86 | 37660 → 443 [ACK] Seq=223 Ack=2857 Win=71168 Len=0 TSval=233609398 TSecr=3691152464 |
| 28 | 10.417914145 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 86 | 37660 → 443 [ACK] Seq=223 Ack=3061 Win=73728 Len=0 TSval=233609399 TSecr=3691152464 |
| 29 | 10.422222709 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 179 | Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message |
| 30 | 10.431818022 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 86 | 443 → 37660 [ACK] Seq=3061 Ack=316 Win=218624 Len=0 TSval=3691152480 TSecr=233609403 |
| 31 | 10.431818523 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TLSv1.2 | 344 | New Session Ticket, Change Cipher Spec, Encrypted Handshake Message |
| 32 | 10.432648471 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 185 | Application Data |
| 33 | 10.432729032 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 372 | Application Data |
| 34 | 10.434149604 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 1514 | [TCP Retransmission] 443 → 37664 [ACK] Seq=1 Ack=223 Win=218624 Len=1428 TSval=2766903391 TSecr=233609398 |
| 35 | 10.434190481 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 98 | 37664 → 443 [ACK] Seq=223 Ack=1429 Win=68096 Len=0 TSval=233609415 TSecr=2766903391 SLE=2857 SRE=3061 |

### ICMP Filter

icmp

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 8559 | 32.105916908 | 192.168.29.7 | 192.168.29.1 | ICMP | 71 | Destination unreachable (Port unreachable) |
| 23273 | 125.386878580 | 192.168.29.7 | 192.168.29.1 | ICMP | 71 | Destination unreachable (Port unreachable) |
| 23517 | 126.744262797 | 192.168.29.1 | 192.168.29.7 | ICMP | 60 | Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 23518) |
| 23518 | 126.744299516 | 192.168.29.7 | 192.168.29.1 | ICMP | 46 | Echo (ping) reply id=0x0000, seq=0/0, ttl=64 (request in 23517) |
| 28615 | 218.877545710 | 192.168.29.7 | 192.168.29.1 | ICMP | 71 | Destination unreachable (Port unreachable) |
| 28644 | 224.384098664 | 192.168.29.1 | 192.168.29.7 | ICMP | 60 | Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 28645) |
| 28645 | 224.384134040 | 192.168.29.7 | 192.168.29.1 | ICMP | 46 | Echo (ping) reply id=0x0000, seq=0/0, ttl=64 (request in 28644) |
| 28808 | 312.517518716 | 192.168.29.7 | 192.168.29.1 | ICMP | 71 | Destination unreachable (Port unreachable) |
| 28886 | 399.712088265 | 192.168.29.1 | 192.168.29.7 | ICMP | 60 | Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 28887) |
| 28887 | 399.712124643 | 192.168.29.7 | 192.168.29.1 | ICMP | 46 | Echo (ping) reply id=0x0000, seq=0/0, ttl=64 (request in 28886) |
| 28905 | 406.237682041 | 192.168.29.7 | 192.168.29.1 | ICMP | 71 | Destination unreachable (Port unreachable) |
| 28955 | 456.083854554 | 192.168.29.1 | 192.168.29.7 | ICMP | 60 | Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 28956) |
| 28956 | 456.083893347 | 192.168.29.7 | 192.168.29.1 | ICMP | 46 | Echo (ping) reply id=0x0000, seq=0/0, ttl=64 (request in 28955) |
| 28997 | 496.233783849 | 192.168.29.1 | 192.168.29.7 | ICMP | 60 | Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 28998) |
| 28998 | 496.233823393 | 192.168.29.7 | 192.168.29.1 | ICMP | 46 | Echo (ping) reply id=0x0000, seq=0/0, ttl=64 (request in 28997) |
| 29005 | 501.121426008 | 192.168.29.7 | 192.168.29.1 | ICMP | 71 | Destination unreachable (Port unreachable) |
| 29054 | 551.733547003 | 192.168.29.1 | 192.168.29.7 | ICMP | 60 | Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 29055) |
| 29055 | 551.733581979 | 192.168.29.7 | 192.168.29.1 | ICMP | 46 | Echo (ping) reply id=0x0000, seq=0/0, ttl=64 (request in 29054) |
| 29084 | 592.943230220 | 192.168.29.1 | 192.168.29.7 | ICMP | 60 | Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 29085) |
| 29085 | 592.943270947 | 192.168.29.7 | 192.168.29.1 | ICMP | 46 | Echo (ping) reply id=0x0000, seq=0/0, ttl=64 (request in 29084) |
| 29091 | 595.628965730 | 192.168.29.7 | 192.168.29.1 | ICMP | 71 | Destination unreachable (Port unreachable) |
| 29146 | 649.464281656 | 192.168.29.1 | 192.168.29.7 | ICMP | 60 | Echo (ping) request id=0x0000, seq=0/0, ttl=64 (reply in 29147) |
| 29147 | 649.464318495 | 192.168.29.7 | 192.168.29.1 | ICMP | 46 | Echo (ping) reply id=0x0000, seq=0/0, ttl=64 (request in 29146) |

Combining Filters

tcp || icmp

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 13 | 10.392455227 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 94 | 37660 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=233609373 TSecr=0 WS=512 |
| 14 | 10.392645999 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 94 | 37664 → 443 [SYN] Seq=0 Win=64800 Len=0 MSS=1440 SACK_PERM TSval=233609373 TSecr=0 WS=512 |
| 15 | 10.402946384 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 94 | 443 → 37660 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM TSval=3691152451 TSecr=233609373 WS=512 |
| 16 | 10.402946945 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 94 | 443 → 37664 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 SACK_PERM TSval=2766903359 TSecr=233609373 WS=512 |
| 17 | 10.402991428 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 86 | 37660 → 443 [ACK] Seq=1 Ack=1 Win=65024 Len=0 TSval=233609384 TSecr=3691152451 |
| 18 | 10.403011000 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 86 | 37664 → 443 [ACK] Seq=1 Ack=1 Win=65024 Len=0 TSval=233609384 TSecr=2766903359 |
| 19 | 10.403174931 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 308 | Client Hello (SNI=firefox.settings.services.mozilla.com) |
| 20 | 10.403378805 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 308 | Client Hello (SNI=firefox.settings.services.mozilla.com) |
| 21 | 10.412344483 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 86 | 443 → 37664 [ACK] Seq=1 Ack=223 Win=218624 Len=0 TSval=2766903369 TSecr=233609384 |
| 22 | 10.417170689 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TLSv1.2 | 290 | [TCP Previous segment not captured] , Ignored Unknown Record |
| 23 | 10.417172973 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 86 | 443 → 37660 [ACK] Seq=1 Ack=223 Win=218624 Len=0 TSval=3691152462 TSecr=233609384 |
| 24 | 10.417173023 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TLSv1.2 | 2942 | Server Hello, Certificate |
| 25 | 10.417173073 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TLSv1.2 | 290 | Server Key Exchange, Server Hello Done |
| 26 | 10.417336811 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 98 | [TCP Dup ACK 18#1] 37664 → 443 [ACK] Seq=223 Ack=1 Win=65024 Len=0 TSval=233609398 TSecr=2766903369 SLE=2857 SRE=3061 |
| 27 | 10.417856270 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 86 | 37660 → 443 [ACK] Seq=223 Ack=2857 Win=71168 Len=0 TSval=233609398 TSecr=3691152464 |
| 28 | 10.417914145 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 86 | 37660 → 443 [ACK] Seq=223 Ack=3061 Win=73728 Len=0 TSval=233609399 TSecr=3691152464 |
| 29 | 10.422222709 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 179 | Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message |
| 30 | 10.431818022 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 86 | 443 → 37660 [ACK] Seq=3061 Ack=316 Win=218624 Len=0 TSval=3691152480 TSecr=233609403 |
| 31 | 10.431818523 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TLSv1.2 | 344 | New Session Ticket, Change Cipher Spec, Encrypted Handshake Message |
| 32 | 10.432648471 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 185 | Application Data |
| 33 | 10.432729032 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TLSv1.2 | 372 | Application Data |
| 34 | 10.434149604 | 2a04:4e42:19::347 | 2405:201:5008:7c02:… | TCP | 1514 | [TCP Retransmission] 443 → 37664 [ACK] Seq=1 Ack=223 Win=218624 Len=1428 TSval=2766903391 TSecr=233609398 |
| 35 | 10.434190481 | 2405:201:5008:7c02:… | 2a04:4e42:19::347 | TCP | 98 | 37664 → 443 [ACK] Seq=223 Ack=1429 Win=68096 Len=0 TSval=233609415 TSecr=2766903391 SLE=2857 SRE=3061 |